

SSTF

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GNU nano 8.7 sstf.c Modified
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;

    int visited[8] = {0};
    int head = 53;
    int total_movement = 0;

    printf("Order of execution:\n");

    for(int i = 0; i < n; i++) {
        int min = 10000;
        int index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(head - req[j]);

                if(distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total_movement += min;
        head = req[index];

        printf("%d -> ", head);

        if(i % 10 == 9) printf("\n");
    }

    printf("\n\nTotal Head Movement = %d\n", total_movement);

    return 0;
}
```

```
athar@THOR MSYS ~
$ nano sstf.c

athar@THOR MSYS ~
$ gcc sstf.c -o sstf

athar@THOR MSYS ~
$ ./sstf
Order of execution:
41 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 ->

Total Head Movement = 323

athar@THOR MSYS ~
$
```